

Automated EBS Snapshot Creation and Cleanup

Monday, July 13, 2026

5:16 PM

1. EBS Volume Setup

Console Navigation

1. Sign in to the **AWS Management Console**.
2. In the top search bar, type **EC2** and select it.
3. In the left sidebar, under **Elastic Block Store**, click **Volumes**.
4. To use an existing volume: note the **Volume ID** (starts with vol-) from the list.
5. To create a new test volume:
 - Click **Create volume** (top right).
 - **Volume type**: gp3
 - **Size**: 8 GiB
 - **Availability Zone**: pick one matching your test EC2 instance's AZ (e.g., us-east-1a)
 - Scroll to **Tags** → **Add tag** → Key: Name, Value: lambda-backup-test
 - Click **Create volume**.
6. Copy the new **Volume ID** — you'll need it for the Lambda environment variable.

EC2 > Volumes > Create volume

Create volume [Info](#)

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Volume type [Info](#)

General Purpose SSD (gp3)

Size (GiB) [Info](#)

8

Min: 1 GiB, Max: 65536 GiB.

IOPS [Info](#)

3000

Min: 3000 IOPS, Max: 80000 IOPS.

Throughput (MiB/s) [Info](#)

125

Min: 125 MiB, Max: 2000 MiB. Baseline: 125 MiB/s.

Availability Zone [Info](#)

us-east-1a (us-east-1a)

Snapshot ID - optional [Info](#)

Don't create volume from a snapshot

Encryption [Info](#)

Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.

☐ Encrypt this volume

Tags - optional [Info](#)

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Remove

Add new tag

You can add up to 49 more tags.

Snapshot summary [Info](#)

Click refresh to view backup information

The volume type that you select and the tags that you assign determine whether the volume will be backed up by any Data Lifecycle Manager policies.

Cancel

Create volume

Automated EBS Snapshot Creation and Cleanup Page 1

Create a new test volume

aws ec2 create-volume --availability-zone us-east-1a --size 8 --volume-type gp3 --tag-specifications "ResourceType=volume,Tags=[{Key=Name,Value=lambda-backup-test}]"

```
PS D:\Cloud-Devops\Devops-Learn-Projects\Devops-Projects\task2\ebs-snapshot-cleanup> aws ec2 create-volume --availability-zone us-east-1a --size 8 --volume-type gp3 --tag-specifications "ResourceType=volume,Tags=[{Key=Name,Value=lambda-backup-test}]"
{
  "AvailabilityZoneId": "use1-az1",
  "Iops": 3000,
  "Tags": [
    {
      "Key": "Name",
      "Value": "lambda-backup-test"
    }
  ],
  "VolumeType": "gp3",
  "MultiAttachEnabled": false,
  "Throughput": 125,
  "VolumeId": "vol-03fb1ddae64ddfed9",
  "Size": 8,
  "SnapshotId": "",
  "AvailabilityZone": "us-east-1a",
  "State": "creating",
  "CreateTime": "2026-07-13T14:14:25+00:00",
  "Encrypted": false
}
```

```
PS D:\Cloud-Devops\Devops-Learn-Projects\Devops-Projects\task2\ebs-snapshot-cleanup> aws ec2 describe-volumes
{
  "Volumes": [
    {
      "AvailabilityZoneId": "use1-az1",
      "Iops": 3000,
      "Tags": [
        {
          "Key": "Name",
          "Value": "lambda-backup-test"
        }
      ],
      "VolumeType": "gp3",
      "MultiAttachEnabled": false,
      "Throughput": 125,
      "Operator": {
        "Managed": false
      },
      "VolumeId": "vol-03fb1ddae64ddfed9",
      "Size": 8,
      "SnapshotId": "",
      "AvailabilityZone": "us-east-1a",
      "State": "available",
      "CreateTime": "2026-07-13T14:14:25.867000+00:00",
      "Attachments": [],
      "Encrypted": false
    }
  ]
}
```

PS D:\Cloud-Devops\Devops-Learn-Projects\Devops-Projects\task2\ebs-snapshot-cleanup>

2. IAM Role for Lambda

Console Navigation

1. Search bar → **IAM** → open the IAM console.
2. Left sidebar → **Roles** → **Create role**.
3. **Trusted entity type**: AWS service
4. **Use case**: select Lambda → **Next**.
5. On the **Add permissions** page, skip attaching a managed policy for now (you'll add an inline one after creation) → **Next**.
6. **Role name**: LambdaEBSBackupRole → **Create role**.
7. Open the newly created role → **Add permissions** dropdown → **Create inline policy**.

8. Click the **JSON** tab and paste:

The image displays three sequential screenshots from the AWS IAM console, illustrating the steps to create a new role.

Step 1: Select trusted entity

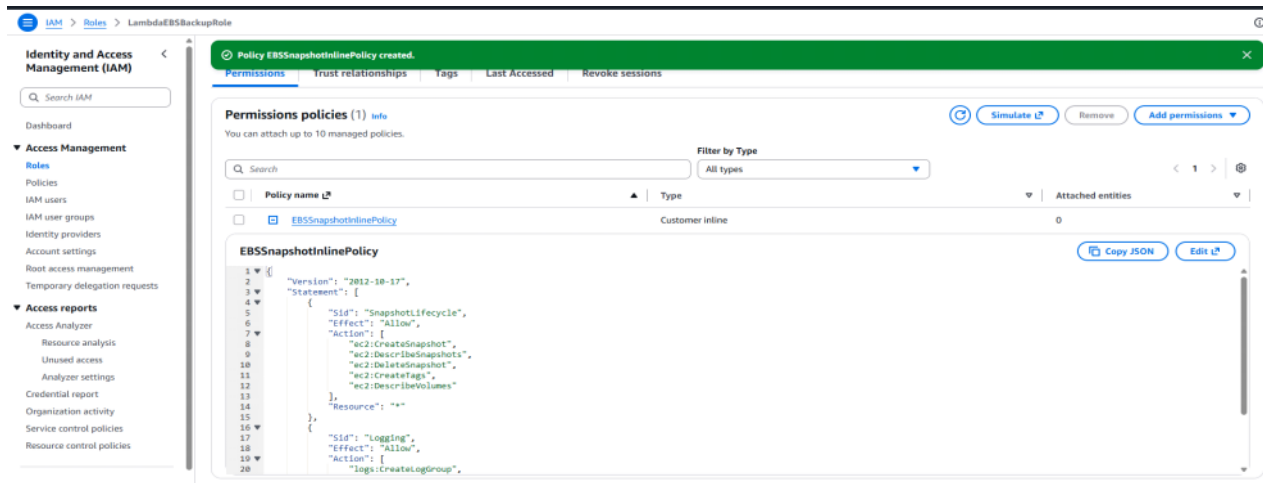
The first screenshot shows the 'Select trusted entity' step. The 'Trusted entity type' is set to 'AWS service'. The 'Use case' is set to 'Lambda'. The 'Service or use case' is set to 'Lambda'. The 'Use case' is set to 'Lambda'.

Step 2: Name, review, and create

The second screenshot shows the 'Name, review, and create' step. The 'Role name' is 'LambdaEBSBackupRole'. The 'Description' is 'Allows Lambda functions to call AWS services on your behalf.'.

Step 3: Permissions

The third screenshot shows the 'Permissions' tab for the created role. The 'Add permissions' button is highlighted.



arn:aws:iam::675134942081:role/LambdaEBSBackupRole

Get the Role ARN

aws iam get-role --role-name LambdaEBSBackupRole --query "Role.Arn" --output text

```
PS D:\Cloud-Devops\Devops-Learn-Projects\Devops-Projects\task2\ebbs-snapshot-cleanup> aws iam get-role --role-name LambdaEBSBackupRole --query "Role.Arn" --output text
arn:aws:iam::675134942081:role/LambdaEBSBackupRole

PS D:\Cloud-Devops\Devops-Learn-Projects\Devops-Projects\task2\ebbs-snapshot-cleanup>
```

3. Create the Lambda Function

Console Navigation

1. Search bar → **Lambda** → open the Lambda console.
2. Click **Create function**.
3. Select **Author from scratch**.
4. **Function name:** EBSBackupCleanup
5. **Runtime:** Python 3.12
6. **Architecture:** x86_64
7. Expand **Change default execution role** → select **Use an existing role** → choose LambdaEBSBackupRole.
8. Click **Create function**.
9. In the **Code source** editor, delete the placeholder code and paste the script below (see full code in the next section).
10. Click **Deploy**.
11. Go to the **Configuration** tab → **General configuration** → **Edit**:
 - **Timeout:** set to 1 min 0 sec
 - Save.
12. Configuration tab → **Environment variables** → **Edit** → **Add environment variable**:
 - Key: VOLUME_ID, Value: vol-0123456789abcdef0 (your actual volume ID)
 - Key: RETENTION_DAYS, Value: 30
 - Save.

Create function [Info](#)

Choose one of the following options to create your function.

☒ **Author from scratch**
Start with a simple Hello World example.

☐ **Use a blueprint**
Build a Lambda application from sample code and configuration presets for common use cases.

☐ **Container image**
Select a container image to deploy for your function.

Basic information

Function name

Enter a name that describes the purpose of your function.

EBSBackupCleanup

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime [Info](#)

Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.12

Permissions

By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. To use a different role, choose **Custom execution role** under **Additional settings**.

Custom settings [Info](#)

Durable execution - [new](#) [?](#)

Build resilient, stateful applications with automatic failure recovery.



EC2 capacity provider - [new](#) [?](#)

Run Lambda on your preferred EC2 instance types instead of Lambda's serverless compute (default).



▼ Additional settings

Customize your function architecture, execution role, HTTP(S) endpoints, tenant isolation mode, VPC, code signing, KMS key, and tags.

Custom settings [Info](#)

Durable execution - [new](#) [?](#)

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▼ Additional settings

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General [Info](#)

ARM64 architecture

Use ARM64 instead of x86_64 (default).



Custom execution role

Use custom execution role instead of new role with basic Lambda permissions (default).



IAM permissions

[LambdaEBSBackupRole](#)

Networking [Info](#)

Function URL

Assign HTTP(S) endpoints.



Tenant isolation mode - [new](#) [?](#)

Guarantee separate execution environments for each tenant invoking the function.



VPC

Connect to a VPC to securely access private resources.



Code
Test
Monitor
Configuration
Aliases
Versions

Configuration
General configuration
Triggers
Permissions
Destinations
Function URL
Environment variables

General configuration [Info](#)

Description

Memory

Ephemeral storage

Timeout

SnapStart

1 min 0 sec

None

Edit

EBSBackupCleanup

Throttle
Copy ARN
Actions

Function overview [Info](#)

Diagram
Template

EBSBackupCleanup

Layers (0)

Add trigger
Add destination

Function ARN

arn:aws:lambda:us-east-1:675134942081:function:EBSBackupCleanup

Description

-

Last modified

5 minutes ago

Export to Infrastructure Composer
Download

Code
Test
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Environment variables

Environment variables (2)

The environment variables below are encrypted at rest with the default Lambda service key.

Q Search Environment variables

Key

Value

RETENTION_DAYS

1

VOLUME_ID

vol-03fb1ddae64ddfed9

Edit

4. EventBridge Weekly Schedule

Console Navigation

1. Search bar → **EventBridge** → open the EventBridge console.
2. Left sidebar → **Rules** → **Create rule**.
3. **Name:** WeeklyEBSBackup
4. **Rule type:** Schedule → **Next**.
5. **Schedule pattern:** select **A fine-grained schedule that runs at a specific time** → choose **Cron-based schedule**.
6. Enter cron expression: 0 3 ? * SUN * (runs every Sunday at 03:00 UTC) → **Next**.
7. **Target type:** AWS service.
8. **Select a target:** Lambda function.

9. **Function:** select EBSBackupCleanup → **Next**.
10. Review and click **Create rule**.
 - EventBridge automatically adds the required invoke permission to the Lambda when created through the console.

Add trigger

Trigger configuration [Info](#)

EventBridge (CloudWatch Events)
aws asynchronous schedule management-tools

Rule
Pick an existing rule, or create a new one.

☒ Create a new rule
☐ Existing rules

Rule name
Enter a name to uniquely identify your rule.

WeeklyEBSBackup

Rule description
Provide an optional description for your rule.

WeeklyEBSBackup

Rule type
Trigger your target based on an event pattern, or based on an automated schedule.

☐ Event pattern
☒ Schedule expression

Schedule expression
Self-trigger your target on an automated schedule using [Cron or rate expressions](#). Cron expressions are in UTC.

rate(10 minutes)

Every 10 minutes execute

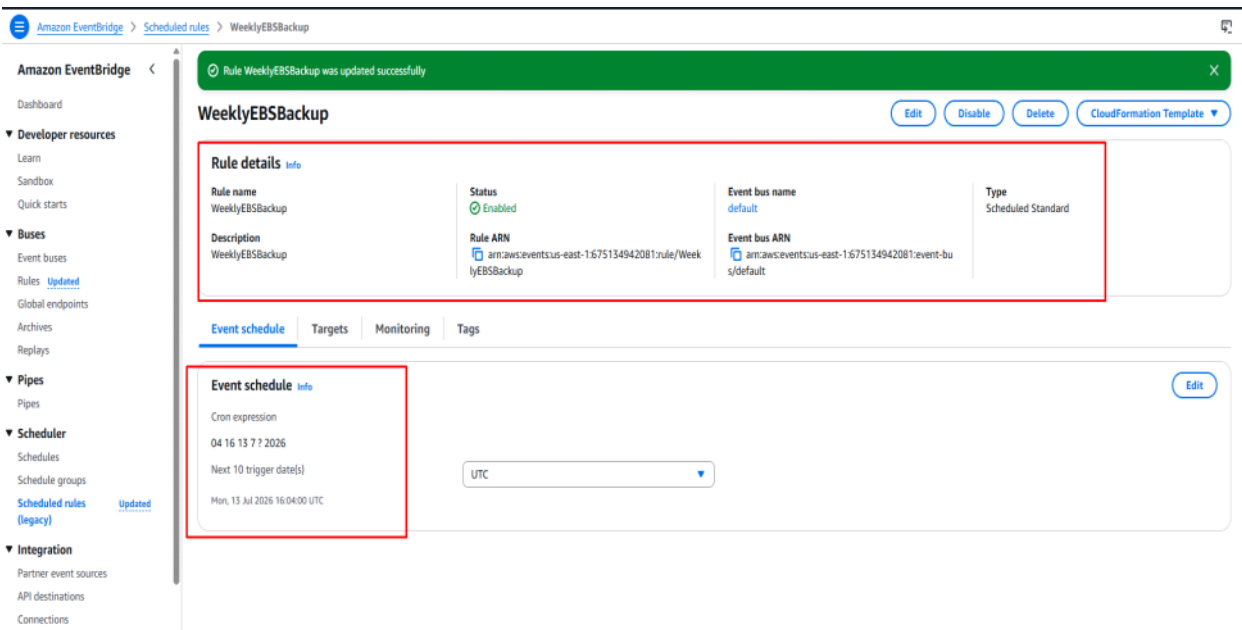
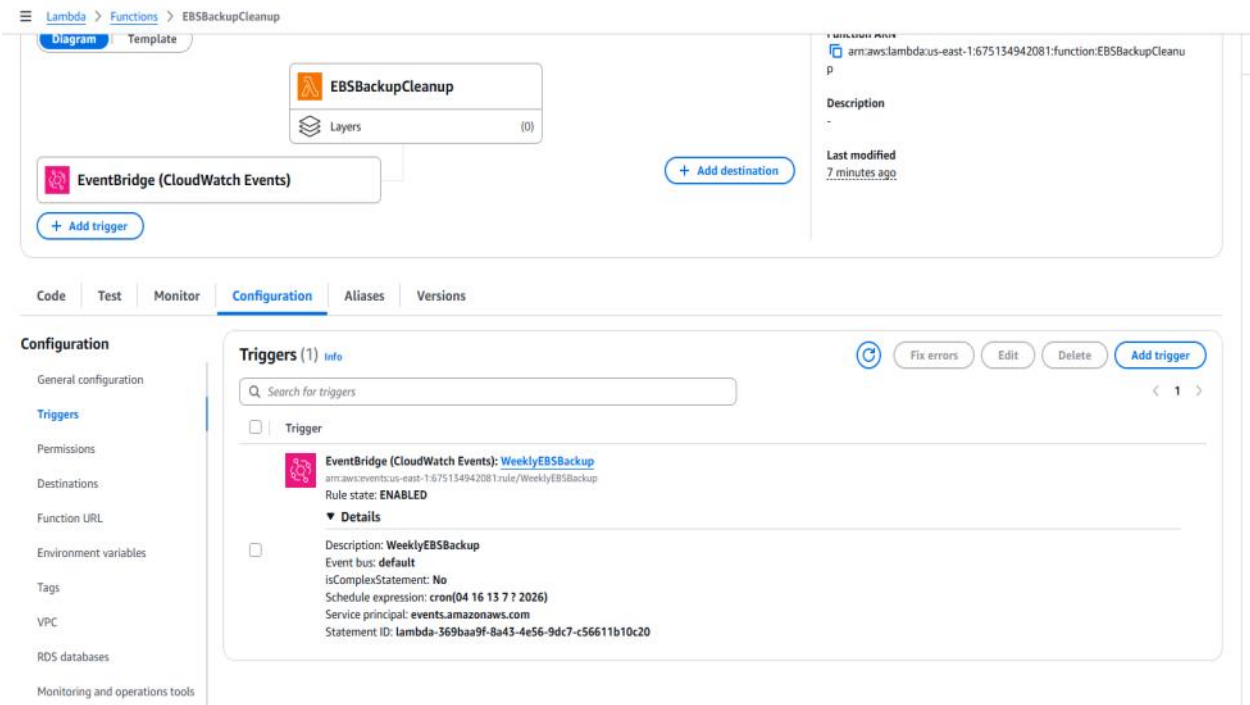
e.g. rate(1 day), cron(0 17 ? * MON-FRI *)

Lambda will add the necessary permissions for Amazon EventBridge (CloudWatch Events) to invoke your Lambda function from this trigger. [Learn more](#) about the Lambda permissions model.

[Cancel](#) [Add](#)

cron(**minutes hours day-of-month month day-of-week year**)

Schedule expression: cron(04 16 13 7 ? 2026)



5. Testing

Console Navigation — Manual Trigger

1. Open the **Lambda** console → select EBSBackupCleanup.
2. Click the **Test** tab.
3. **Event name:** manual-test
4. Leave the default JSON {} (no input needed) → **Save**.
5. Click **Test**.
6. Check the **Execution results** panel at the top — expand **Details** to see the returned JSON (created and deleted snapshot IDs) and the **Log output** with printed statements.



Code **Test** Monitor Configuration Aliases Versions

✓ Executing function: succeeded ([logs](#))

▼ Details

```
{
  "created": "snap-081b7fb70c6f6e856",
  "deleted": []
}
```

Summary

Code SHA-256 M9Sf5eGudBmhwS3A99ofWyikoEblQZbBaQIZ0XM0Le4=	Execution time 3 seconds ago
Function version \$LATEST	Request ID 6b094b84-8368-4976-b33d-ef13e91c4e9c
Duration 1107.98 ms	Billed duration 1559 ms
Resources configured 128 MB	Max memory used 101 MB
Init duration 450.88 ms	

Log output

The area below shows the last 4 KB of the execution log. [Click here](#) to view the corresponding CloudWatch log group.

```
START RequestId: 6b094b84-8368-4976-b33d-ef13e91c4e9c Version: $LATEST
Created snapshot: snap-081b7fb70c6f6e856
Deleted snapshots (0): []
END RequestId: 6b094b84-8368-4976-b33d-ef13e91c4e9c
REPORT RequestId: 6b094b84-8368-4976-b33d-ef13e91c4e9c Duration: 1107.98 ms Billed Duration: 1559 ms Memory Size: 128 MB Max Memory Used: 101 MB Init Duration: 450.88 ms
```

Console Navigation — Verify in EC2 Console

1. Go to **EC2 console** → **Elastic Block Store** → **Snapshots** (left sidebar).
2. Use the search/filter bar: filter by tag CreatedBy = Lambda-Backup.
3. Confirm the new snapshot appears with **Started** timestamp matching your test run and status **completed**.

EC2 Snapshots (1) info

Owned by me Search

Name	Snapshot ID	Full snapshot size	Volume size	Description	Storage tier	Snapshot status	Started	Progress
	snap-081b7fb70c6f6e856	0 B	8 GiB	Automated snapshot of vol...	Standard	Completed	2026/07/13 21:30 GMT+5...	100%

4. To test cleanup: go back to Lambda → **Configuration** → **Environment variables** → temporarily set RETENTION_DAYS to 0 → **Save** → re-run **Test** tab.

Edit environment variables

Environment variables

You can define environment variables as key-value pairs that are accessible from your function code. These are useful to store configuration settings without the need to change function code. [Learn more](#)

Key	Value	
RETENTION_DAYS	0	<button>Remove</button>
VOLUME_ID	vol-03fb1ddae64ddfed9	<button>Remove</button>
<button>Add environment variable</button>		

Encryption configuration

[Cancel](#)

[Save](#)

Executing function: succeeded [logs](#)

Details

```
{
  "created": "snap-0071e5d207b19f99",
  "deleted": [
    "snap-0044331c440709a79",
    "snap-0071e5d207b19f99",
    "snap-001b7f070c6f0e056"
  ]
}
```

Summary

Code SHA-256
M9SfSeGud8mhS3A99oWYjKoEblQZbBaQZDlXMDLe4+

Execution time
1 minute ago

Function version
\$LATEST

Request ID
bb911982-6e5c-448d-bcd3-5440422b3759

Duration
1694.68 ms

Billed duration
2301 ms

Resources configured
128 MB

Max memory used
102 MB

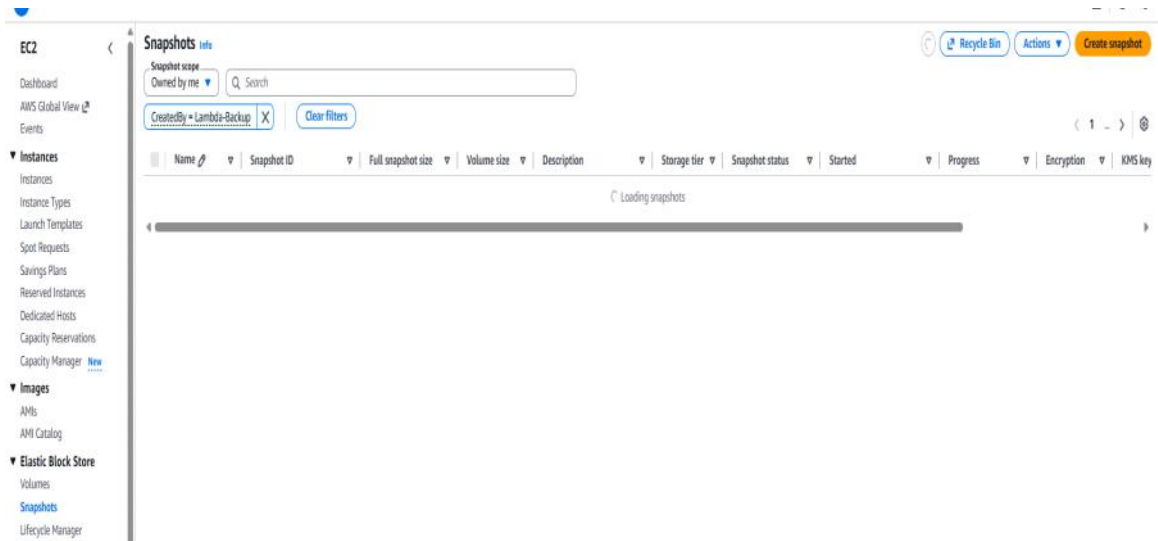
Init duration
605.81 ms

Log output

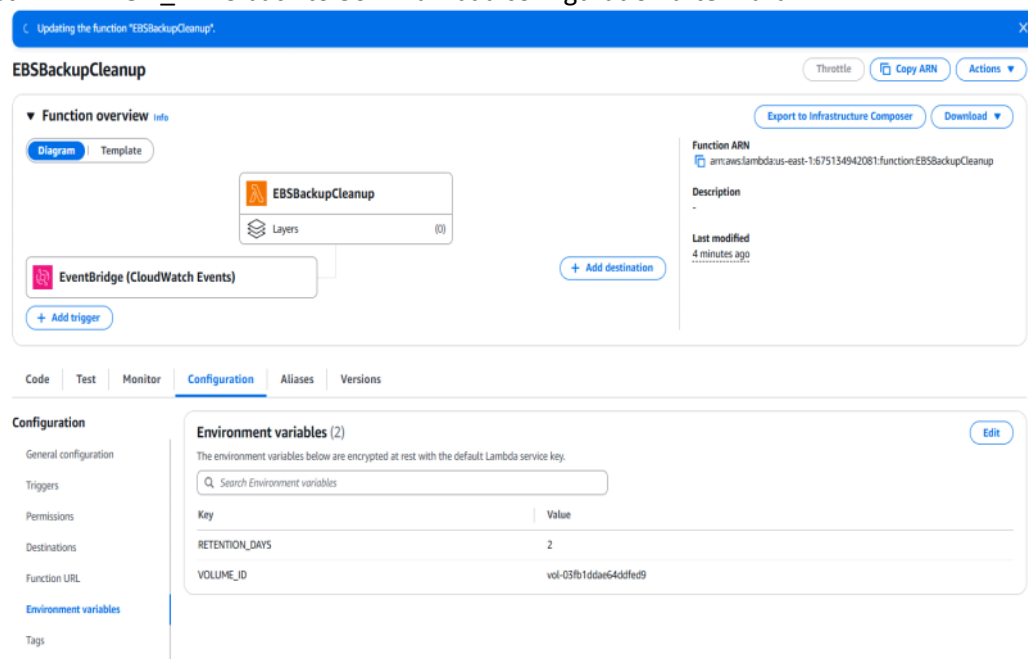
The area below shows the last 4 KB of the execution log. [Click here](#) to view the corresponding CloudWatch log group.

```
START RequestId: bb911982-6e5c-448d-bcd3-5440422b3759 Version: $LATEST
Created snapshot: snap-0071e5d207b19f99
Deleted snapshots (3): ["snap-0044331c440709a79", "snap-0071e5d207b19f99", "snap-001b7f070c6f0e056"]
END RequestId: bb911982-6e5c-448d-bcd3-5440422b3759
REPORT RequestId: bb911982-6e5c-448d-bcd3-5440422b3759 Duration: 1694.68 ms Billed Duration: 2301 ms Memory Size: 128 MB Max Memory Used: 102 MB Init Duration: 605.81 ms
```

5. Refresh the Snapshots list in EC2 console — older tagged snapshots should disappear.



6. Reset RETENTION_DAYS back to 30 in Lambda configuration afterward.



Lambda > Functions > EBSBackupCleanup

```

{
  "created": "snap-0ef62938bd298339",
  "deleted": []
}

```

Summary

Code SHA-256 M9SF5eGudBmHxS3A99ofWyik0EblQZbBaQIZ0XMDLe4+	Execution time 2 minutes ago
Function version \$LATEST	Request ID 573295cf-3810-4827-91be-11a18fe28f61
Duration 950.87 ms	Billed duration 1516 ms
Resources configured 128 MB	Max memory used 99 MB
Init duration 564.56 ms	

Log output

The area below shows the last 4 KB of the execution log. [Click here](#) to view the corresponding CloudWatch log group.

```

START RequestId: 573295cf-3810-4827-91be-11a18fe28f61 Version: $LATEST
Created snapshot: snap-0ef62938bd298339
Deleted snapshots: []
END RequestId: 573295cf-3810-4827-91be-11a18fe28f61
REPORT RequestId: 573295cf-3810-4827-91be-11a18fe28f61 Duration: 950.87 ms Billed Duration: 1516 ms Memory Size: 128 MB Max Memory Used: 99 MB Init Duration: 564.56 ms

```

Change Cron run every 10 min for Testing

Code | **Test** | **Monitor** | **Configuration** | **Aliases** | **Versions**

Configuration

General configuration

Triggers

Permissions

Destinations

Function URL

Environment variables

Tags

VPC

RDS databases

Monitoring and operations tools

Triggers (1) Info

Search for triggers

Trigger

EventBridge (CloudWatch Events): WeeklyEBSBackup
arn:aws:events:us-east-1:675134942081:rule/WeeklyEBSBackup
Rule state: **ENABLED**

Details

Description: WeeklyEBSBackup
Event bus: default
isComplexStatement: No
Schedule expression: rate(10 minutes)
Service principal: events.amazonaws.com
Statement ID: lambda-369baa9f-8a43-4e56-9dc7-c56611b10c20

Console Navigation — View Logs

1. Lambda console → EBSBackupCleanup → **Monitor** tab → **View CloudWatch logs**.
2. Click the most recent **Log stream** to see the printed created/deleted snapshot IDs.

aws | Search | [Alt+S] | United States (N. Virginia) | poweruser

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

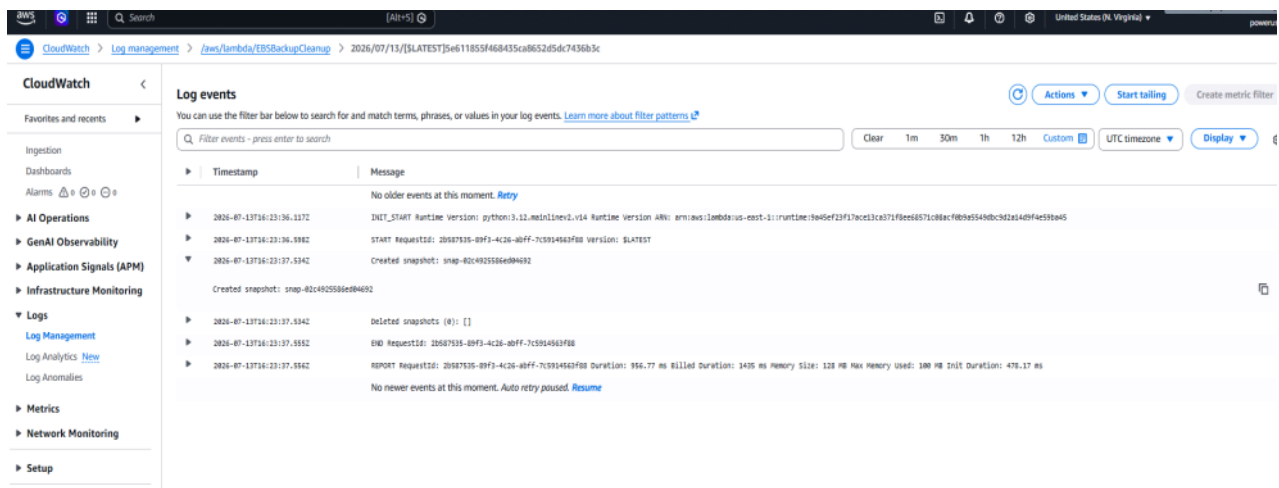
Snapshots (3) Info

Snapshot scope: Owned by me

CreatedBy = Lambda-Backup

Clear filters

Name	Snapshot ID	Full snapshot size	Volume size	Description	Storage tier	Snapshot status	Started	Progress	Encryption	KMS key
→ snap-02c4925586ed04692	-	-	8 GiB	Automated snapshot of vol...	Standard	Pending	2026/07/13 21:53 GMT+5...	99%	Not encrypted	-
→ snap-0002938bd298339	0 B	8 GiB	8 GiB	Automated snapshot of vol...	Standard	Completed	2026/07/13 21:39 GMT+5...	100%	Not encrypted	-
→ snap-0cc4005596a36611a	0 B	8 GiB	8 GiB	Automated snapshot of vol...	Standard	Completed	2026/07/13 21:43 GMT+5...	100%	Not encrypted	-

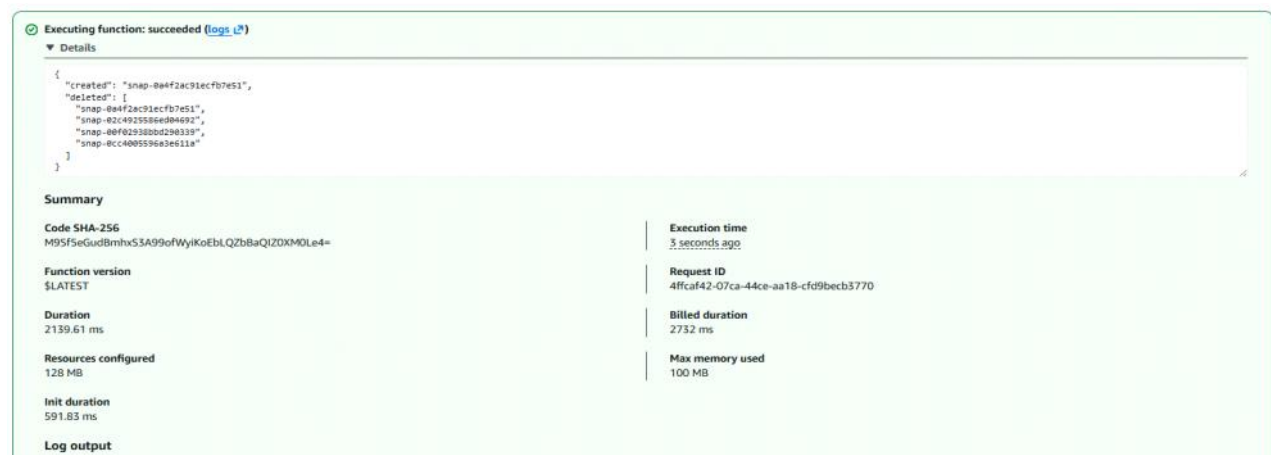


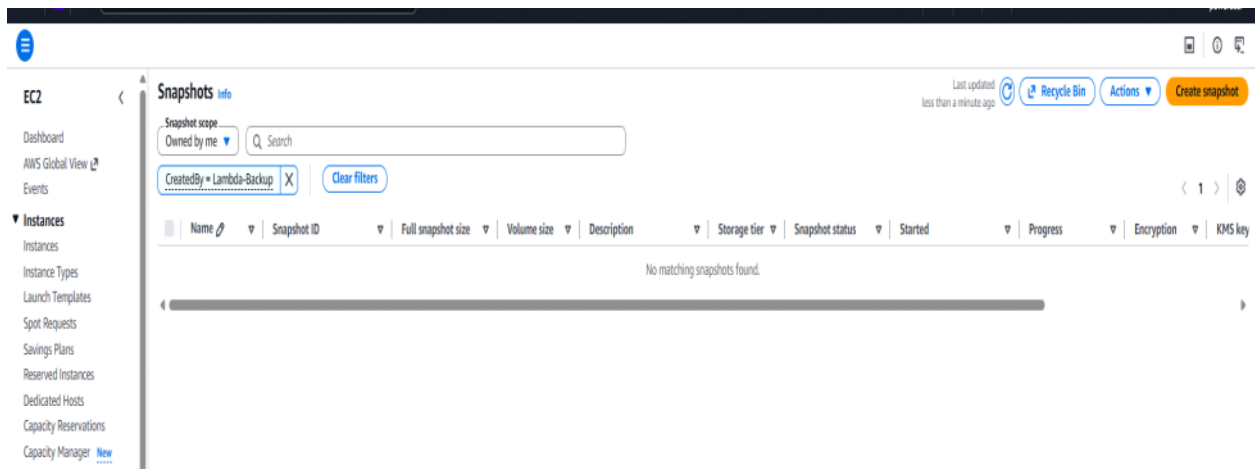
CLI Equivalent

`aws logs tail /aws/lambda/EBSBackupCleanup --follow`

```
PS D:\Cloud-Devops\Devops-Learn-Projects\Devops-Projects\task2\lebs-snapshot-cleanup> aws logs tail /aws/lambda/EBSBackupCleanup --follow
2026-07-13T16:13:35.993000+00:00 2026/07/13/[LATEST]1bf8e687c08acfb09a540db9c9d2a14d9f4e59ba45 INIT_START Runtime Version: python:3.12.mainline.v14 Runtime Version ARN: arn:aws:lambda:us-east-1::runtime:9a45ef23f1ace13ca371f8e687c08acfb09a540db9c9d2a14d9f4e59ba45
2026-07-13T16:13:37.065000+00:00 2026/07/13/[LATEST]1bf8e687c08acfb09a540db9c9d2a14d9f4e59ba45 Created snapshot: snap-0cc4005596a3e611a
2026-07-13T16:13:37.065000+00:00 2026/07/13/[LATEST]1bf8e687c08acfb09a540db9c9d2a14d9f4e59ba45 Deleted snapshots (0): []
2026-07-13T16:13:37.072000+00:00 2026/07/13/[LATEST]1bf8e687c08acfb09a540db9c9d2a14d9f4e59ba45 END RequestId: eaefde9b-a975-4b05-ae3d-15f054190237
2026-07-13T16:13:37.072000+00:00 2026/07/13/[LATEST]1bf8e687c08acfb09a540db9c9d2a14d9f4e59ba45 REPORT RequestId: eaefde9b-a975-4b05-ae3d-15f054190237 Duration: 1079.00 ms Billed Duration: 1079 ms Memory Size: 128 MB Max Memory Used: 99 MB
2026-07-13T16:23:36.117000+00:00 2026/07/13/[LATEST]5e611855f468435ca8652d5dc7436b3c INIT_START Runtime Version: python:3.12.mainline.v14 Runtime Version ARN: arn:aws:lambda:us-east-1::runtime:9a45ef23f1ace13ca371f8e687c08acfb09a540db9c9d2a14d9f4e59ba45
2026-07-13T16:23:36.598000+00:00 2026/07/13/[LATEST]5e611855f468435ca8652d5dc7436b3c START RequestId: 2b587535-89f3-4c26-abff-7c5914563f88 Version: $LATEST
2026-07-13T16:23:37.534000+00:00 2026/07/13/[LATEST]5e611855f468435ca8652d5dc7436b3c Created snapshot: snap-02c4025586e404692
2026-07-13T16:23:37.534000+00:00 2026/07/13/[LATEST]5e611855f468435ca8652d5dc7436b3c Deleted snapshots (0): []
2026-07-13T16:23:37.556000+00:00 2026/07/13/[LATEST]5e611855f468435ca8652d5dc7436b3c END RequestId: 2b587535-89f3-4c26-abff-7c5914563f88
2026-07-13T16:23:37.556000+00:00 2026/07/13/[LATEST]5e611855f468435ca8652d5dc7436b3c REPORT RequestId: 2b587535-89f3-4c26-abff-7c5914563f88 Duration: 956.77 ms Billed Duration: 1435 ms Memory Size: 128 MB Max Memory Used: 100 MB Init Duration: 478.17 ms
```

Set to again `>>> RETENTION_DAYS=0`





6. Lambda vs. AWS Data Lifecycle Manager (DLM)

DLM Console Navigation (for comparison/reference):

1. EC2 console → left sidebar (bottom, under **Elastic Block Store**) → **Lifecycle Manager**.
2. **Create lifecycle policy** → choose **EBS snapshot policy**.
3. Select target volumes by tag, set schedule frequency and retention count/days, assign an IAM role (DLM has an AWS-managed service-linked role option).
4. No code required — this is DLM's core appeal for simple schedules.